Movie Recommendation System using Hybrid NCF with LightGCN Framework

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***Abstract*— This paper presents a hybrid film recommendation system that utilizes Neural Collaborative Filtering (NCF), Lightweight Graph convolution Networks (LightGCN) model, and comprehensive side features of user and item to increase the precision of top-K movie recommendations over large datasets. Traditional NCF models do an excellent job of modelling the nonlinear interactions between users and movies however NCFs have difficulties with sparse rating matrices and LightGCNs' modelling of the context is done through graph structures which often miss contextual signals. To overcome the limitations of NCFs and LightGCNs, the proposed model combines the use of the Generalised Matrix Factorisation (GMF) & Multi-Layer Perception (MLP) components of NCF, with the propagation methodology of LightGCN in order to leverage advanced user-item interactions and structural neighbourhood embeddings. Additionally, auxiliary information such as demographic information about the user, and genre information of the item have been included to enhance representation quality and help to alleviate cold-start issues. Experiments conducted using the MovieLens-1M dataset demonstrated that the proposed hybrid system outperforms the independent baselines for HR@10 and NDCG@10, and begins to converge earlier and generalises better across the sparsest segments of users. Overall, these findings support the integration of deep modelling of interactions, graph-based modelling of learning and fusion of side information to produce an advanced recommender system.**

I. Introduction

In this vast agglomeration of data and information that subsist on the internet. Users or the customers are always disoriented about the digital content, products, and the services. As online platforms boom connecting apropos data that full fill the user's precedence has become increasingly backbreaking and it has become a major knot for all. Recommender systems are now to be considered a key means of information filtering and providing customized recommendations in a broad spectrum of applications such as e-commerce, entertainment, social media, and learning systems[1]. Among different formulation, top-P recommendation has become majorly espoused evaluation setting, achingly in large-scale environments where ranking accuracy directly impacts users satisfaction.

traditional collaborative filtering approaches, involve memory-based and early model-based methods, provided foundation for personalized ranking but struggle with sparsity, scalability, and the cold start-problem[2].With involvement of deep learning advance model, Neural Collaborative Filtering (NCF), a non-linear interaction modelling was establish that further accentuated predictive performance[3].However pure neural models often fail to fully exploit the graph-structured nature of user-item relationships.

Recent graph-based approaches, especially Light Graph Convolutional Networks (LightGCN) have done a powerful performance by simplifying propagation operations and preserving collaborative signals without any needless transformations [4]. But graph-only models often lack the resilience of leveraging additional contextual information, while neural-only model may not capture high-order connectivity with in user-item graphs.

To fill these gaps, we proposed a Hybrid Recommendation Framework by combining Neural Collaborative Filtering, side feature embeddings, and LightGCN-based graph propagation. The model is trained on the widely used MovieLens-1M dataset, standardized comparison with existing methods. By unifying deep neural interactions, graph-based collaborative signals, and auxiliary side information, the prosed model aims to deliver more accurate Top-P movie recommendations while mitigating sparsity and improvise generalization

The results of this study might have a major influence on recommendation since they offer a trustworthy method of recommendation systems. Moreover, as the dataset expands and model designs get better, real time recommendations become ever more possible. In the end, this work aims to support the continuous initiatives among social media and online platform to use the recommender system for making the recommendations more accurate.

II. Related Work

Sun et. al [5] introduced the first bidirectional Transformer encoder BERT4Rec for modeling user interaction sequences, by using a Cloze task to predict masked items in a sequence by

conditioning on both left and right context. The proposed model outperforms the traditional deep learning method and achieves the performance on Movielens ML-1m dataset with HR@10 ≈ 0.6788 and NDCG@10 ≈ 0.4631. However, the study was done on various datasets for comparison.

Li et. al [6] introduced DUALRec, a hybrid recommendation framework combining an LSTM-based sequential model with a fine-tuned large language model (LLM) so that it captures both temporal dynamics and semantic context in user movie preferences. DUALRec outperforms traditional baselines by effectively uniting chronological behavioral signal with language -based semantic modeling. On MovieLens-1m, DUALRec achieve the highest HR@5 ≈ 0.2214 and NDCG@5 ≈ 0.2078 among compared models.

Chen et. al [7] introduced J-NCF that proposed joint neural architecture that integrate deep feature learning and deep interaction modeling. They test the model's robustness under different data sparsity conditions and user activity levels. On MovieLens-1M, J-NCF achieve HR@ improvements of +10.81% over baselines and NDCG@10 improvements of +14.24% that are approximately 0.7377 and 0.4822.

Zheng, et. al [8], Lighter Graph Convolutional Networks (Lighter-X) proposed a parameter-efficient GCN framework for recommendation, reducing dense embeddings via low rand random projections while preserving performance. In this, Experiments on large-scale datasets show that Lighter-X achieves comparable or better HR@10 and NDCG@10 than standard LightGCN. It achieve the NDCG@10 ≈ 0.1538.

He et. al. (2017) proposed Neural Collaborative Filtering (NCF) which replace the traditional inner product in matrix factorization with a neural network to model complex user-item interactions. The framework includes GMF, MLP, and a combined NeuMF architecture for capturing both linear and non-linear patterns. They done their experiments on two datasets-MovieLens and Pinterest for Comparison. With NeuMF(pre-training), on MovieLens they report HR@ ≈ 0.726 and NDCG@10 ≈ 0.445. NCF established a strong baseline for deep learning- based recommendation systems.

III. Methodology

A. Data Collection

In this paper we use the MovieLens -1M dataset. This dataset containing 1 million ratings from 6040 users on 3952 movies. The dataset includes three datasets – Ratings are on 5-point scale, and each user has rated atleast 20 movies. The dataset includes users demographics (gender, age, occupation) and movie metadata (titles and genres), enabling hybrid recommendation models.
Its sparsity (~4% of all possible ratings) (fig.1) and timestamp information make it suitable for both Top-K recommendation tasks and sequential or temporal modeling.

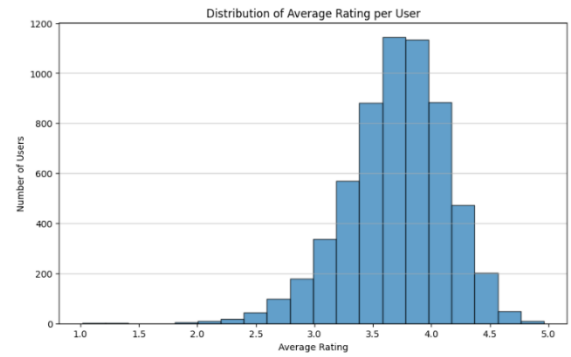

fig.1 Distribution of Average Rating per user

B. Data Preprocessing

Before training the Hybrid NCF model. The dataset need to preprocessed and prepared for the training according to model. The preprocessing includes the following steps-

1. ***Rating Binarization*** – ratings greater than or equal to 4 were considered positive interactions, while other were treated a negative interaction (fig2). A binary label column was created:

$$label = \begin{cases} 1, if\ ratings \geq 4 \\ 0, otherwise \end{cases}$$

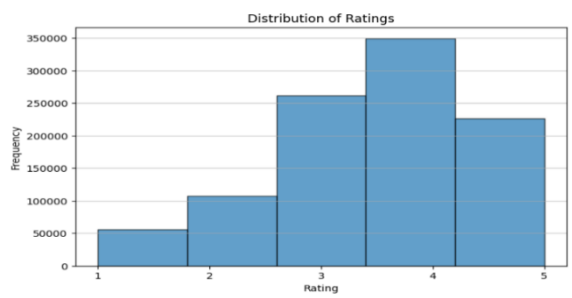

fig.2 Distribution of users per rating

2. **User Features** – User gender was encoded as a binary variable (F=1, M=0). Age was normalized to the [0,1] range.

3. Movie Features – Represent movie genres as multi-hot vectors across all genres categories. Each movie could have multiple genres, resulting in a binary vector indicating the presence of each genre.

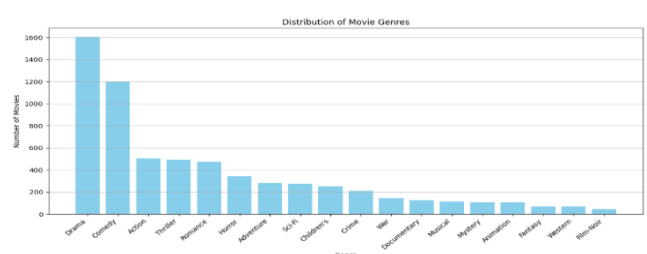

fig3. Movie Genre distributions

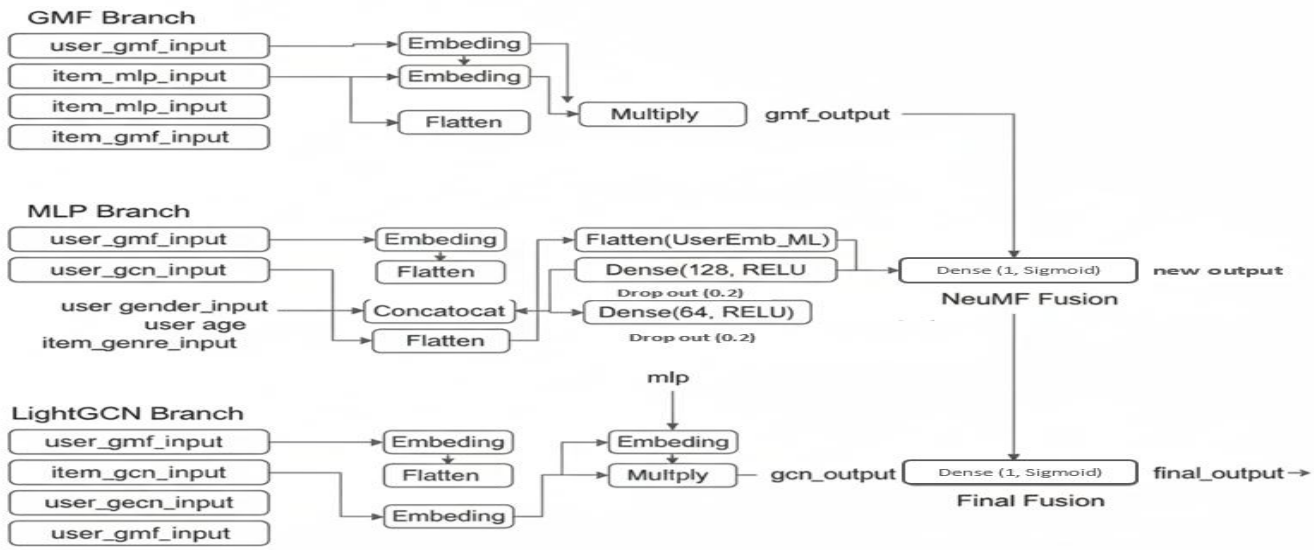

fig.4 Proposed Model Architecture

4. ***Negative Sampling*** – To handle the implicit feedback nature of the dataset, we employed negative sampling. For each positive user-item interaction, 40 negative samples (items not interacted by user) were randomly selected, resulting in a balanced dataset for model training and prevent overfitting.

5.Train-Validation Split– The preprocessed interactions were split into training and validation sets using 80:20 ratio.

6.Side Features – Side features like – gender, age, and genre vectors were aligned with the corresponding user-item pairs.

C. *Proposed Model Architecture*

We proposed a hybrid Neural Collaborative Filtering (NCF) model with LightGCN integration and side information (fig.4), designed to leverage user-item interactions and auxiliary features.

- **Inputs:**
  - User IDs for GMF, MLP, and LightGCN branches
  - Item IDs for GMF, MLP, and LightGCN branches
  - Side features: gender, age, genre vector
- **Embedding Layers:** Separate embeddings for GMF, MLP, and LightGCN branches with L2 regularization (1e-6).
- **MLP Layers:** Dense layers of sizes 128 → 64 → 32 with ReLU activations and dropout of 0.2.
- **Fusion:** Concatenation of GMF, MLP, and LightGCN outputs, followed by a dense layer with sigmoid activation to generate the final prediction.
- **Optimization:** AdamW with a learning rate of 0.0005 and weight decay of 1e-4. As a loss function binary cross-entropy was used.

D. *Training Strategy*

The Hybrid NCF model was compiled and trained using the following configurations:

- **Optimizer**: Adam optimizer, which has an adaptive learning rate
- **Loss Function**: Binary cross entropy loss suitable in binary class classification issues.
- **Metrics**: Loss was tracked during training and evaluation.
- **Negative Sampling per Positive:** 40 negative items sampled for each positive interaction.

Using a batch size of 1024, the model underwent 50 epochs of training with early stopping (patience=5) based on validation loss to prevent overfitting. The 80% of dataset is used for training and 20% dataset is used for validation.

E. Evaluation Metrics

To measure model performance, we implement several metrics:

1.Hit Ratio (HR@P) – Hit Ratio is used to determine the presence of the ground-truth positive item in the top-P items that were accurately predicted by the model. It is defined as:

$$HR@P = \begin{cases} 1, if\ the\ positve\ item\ is\ in\ top - P \\ 0, otherwise \end{cases}$$

The final HR@P value is averaged across all users. A higher HR indicates stronger retrieval capability.

2.Normalized Discounted Cumulative Gain (NDCG@P) - NDCG@P evaluates ranking quality by giving higher importance to items appearing at higher positions in the top-P list. It is defined as:

$$NDCG@P = \frac{1}{\log_2(rank + 2)}$$

where *rank* is the zero-based position of the ground-truth item.

NDCG penalizes models that place relevant items lower in the ranking, making it a stronger indicator of ranking effectiveness.

3. Recall@P

Recall@P measures the proportion of relevant items retrieved in the top-P recommendations. In the evaluation setup, each user has exactly one relevant item, thus:

$$recall@P = HR@P$$

However, we still report recall@P for standardization with related research.

4. Loss Metrics

During training, we track:

- Binary Cross-Entropy Loss (Training Loss)
- Validation Loss

Validation loss is used for early stopping to prevent overfitting.

For each user, a set of 99 randomly sampled negative items is combined with the test item. This evaluation approach provides a realistic approximation of top-K recommendation performance and ensures fair comparison with state-of-the-art baseline models.

IV. RESULT AND ANALYSIS

This section provides the performance of proposed Movie Recommender System model for top-P movies. Loss. Hit Ratio, NDCG and recall to show the evaluation how effectively the model can recommend top-P movies.

A. Training and Validation Loss

The model demonstrates consistently high performance across the training epochs (Fig.6). Training accuracy increases sharply during the initial epochs, rising from approximately **85.6%** in epoch 0 to around **92.1%** by epoch 2. Beyond this point, the improvement stabilizes, reaching a peak of roughly **92.4%** at epoch 6.

Validation accuracy is also positively moving with a steady increase of 90.9 to approximately 91.7 during epochs 0 and 1 respectively, but steadily moving towards 92.2 after 6 epochs. The fact that the training and validation accuracy curves are very close implies that the model is generalized and shows no severe overfitting. The fact that the two curves meet on the common range of accuracy also increases the stability and strength of the learning process.

B. Training and Validation Accuracy

The loss curves (Fig.5) complement the accuracy trends. Training loss drops sharply from **0.095** to **0.078** in the first epoch and reaches its minimum around **0.076** at epoch 2. From epochs 3 to 6, the training loss shows a slight upward drift but remains low and stable, ending at approximately **0.078**.

Validation loss decreases in the initial phase, reaching its minimum at epoch 1 (**0.077**), followed by a gradual increase up to **0.080** at epoch 6. Despite this mild rise in validation loss, the gap between training and validation loss remains small throughout, indicating only minimal overfitting. This behavior is consistent with a model that has learned the underlying data distribution effectively without substantial degradation in generalization performance.

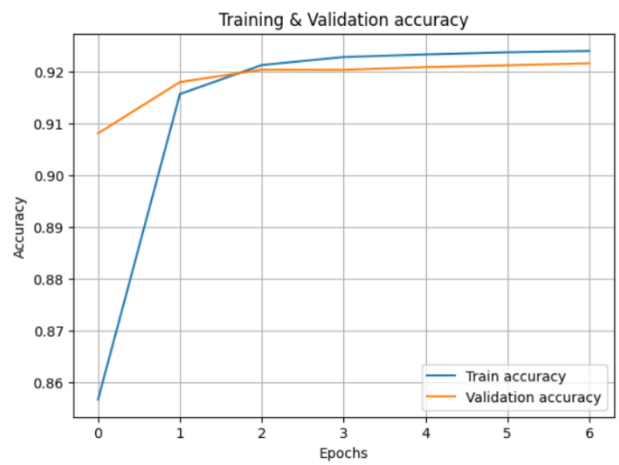

fig.5 Training and Validation Accuracy

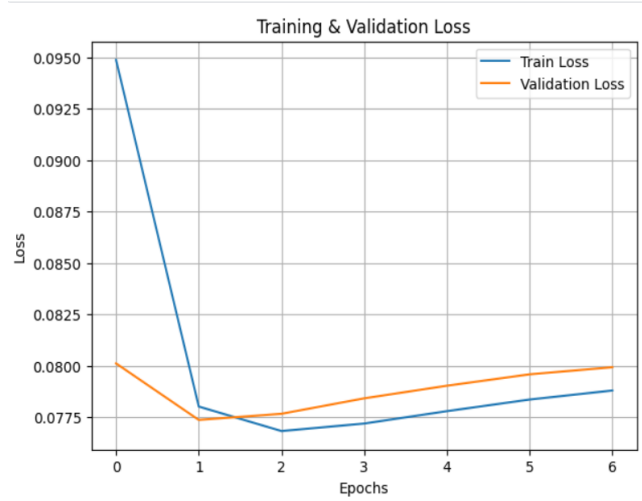

fig.6 Training and Validation Loss

C. Comparison with other models

The performance measures of many deep learning models for movies recommender system are presented in the table I. Five models comprise it: LightGCN, BERT4Rec, NCF, J-NCF and proposed Hybrid + LightGCN Model (Table1). Performance of every model is assessed using Hit Ratio (HR@10) and NDCG@10 in addition to perceptive analysis of their success. While LightGCN and J-NCF followed closely, the suggested Hybrid NCF with LightGCN Model attained the best HR@10 at 0.801, proving minimum overfitting. Though with slower training times and somewhat reduced accuracy, BERT4Rec and NCF performed rather well. These outcomes offer insightful analysis of model choice for tasks involving top-P movies recommendations.

Models	***HR@10***	***NDCG@10***
LightGCN	0.729	0.415
BERT4Rec	0.678	0.463
NCF	0.726	0.445
J-NCF	0.737	0.482
Proposed	*0.801*	*0.596*

Table1 *- Training and Validation Loss*

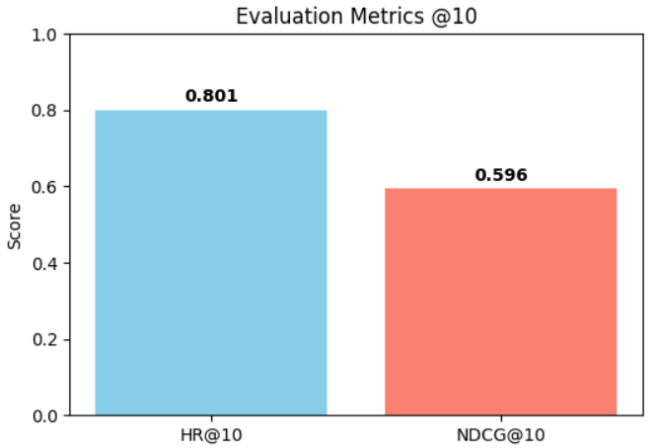

fig.7 Final metrics performance

V. CONCLUSION

In this experiment, we investigated neural network architecture of collaborative filtering using side features. We came up with a broad structure and suggested three instantiations of such a structure, namely Hybrid NCF, LightGCN, and Side Features, which model user-item interactions differently. Our structure is only the combinations of other model with the aim of filling the gap between such models and attempt to minimize their limitations. The score of 0.801 hit rate and 0.596 NDCG indicate that this work is successful in classifying top movies with the help of the hybrid model based on the rating criterion and the side features criteria. The model performed better than many other deep learning structures. This is a better way of making recommendations since it integrates other models hence perhaps making better recommendations. However this study is not the limit here, there will be much more experiments in the future to enhance recommendations on the same.

VI. REFERENCE

[1] Shapira, B., Rokach, L., & Ricci, F. (2022). Recommender systems handbook.

[2] Resnick, P., & Varian, H. R. (1997). Recommender systems. *Communications of the ACM*, *40*(3), 56-58.

[3] He, X., Liao, L., Zhang, H., Nie, L., Hu, X., & Chua, T. S. (2017, April). Neural collaborative filtering. In *Proceedings of the 26th international conference on world wide web* (pp. 173-182).

[4] He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., & Wang, M. (2020, July). Lightgcn: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval* (pp. 639-648).

[5] Sun, F., Liu, J., Wu, J., Pei, C., Lin, X., Ou, W., & Jiang, P. (2019, November). BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. In *Proceedings of the 28th ACM international conference on information and knowledge management* (pp. 1441-1450).

[6] Li, Y., & Grasman, R. (2025). DUALRec: A Hybrid Sequential and Language Model Framework for Context-Aware Movie Recommendation. *arXiv preprint arXiv:2507.13957*.

[7] Chen, W., Cai, F., Chen, H., & Rijke, M. D. (2019). Joint neural collaborative filtering for recommender systems. *ACM Transactions on Information Systems (TOIS)*, *37*(4), 1-30.

[8] Zheng, Y., Wei, Z., Liu, J., de Hoog, F., Chen, X., Ye, Y., & Huang, J. Lighter Graph Convolutional Networks for Recommendation.